

1993 HKCEE MATHS Paper II

1 If $f(x) = 10^{2x}$, then $f(4y) =$

- A. 10^{4y}
 B. 10^{2+4y} D. 40^y
 C. 10^8 E. 40^{2y}

2 If $s = \frac{n}{2}[2a + (n-1)d]$, then $d =$

- A. $\frac{2(s-an)}{n(n-1)}$
 B. $\frac{2(s-an)}{n-1}$ D. $\frac{as-n}{a(n-1)}$
 C. $\frac{s}{n(n-1)}$ E. $\frac{4(s-an)}{n(n-1)}$

3 Simplify $(x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1)$

- A. $x^4 + 1$
 B. $x^4 - x^2 + 1$
 C. $x^4 + x^2 + 1$
 D. $x^4 - 3x^2 - 2\sqrt{3}x - 1$
 E. $x^4 + \sqrt{3}x^3 - 2\sqrt{3}x^2 + \sqrt{3}x + 1$

4 Simplify $\frac{\sqrt{b}}{\sqrt{a}-\sqrt{b}} + \frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}$

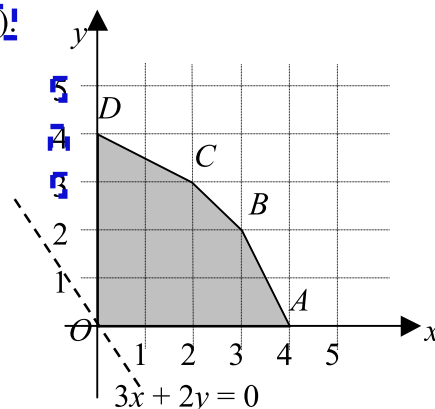
- A. $\frac{1}{\sqrt{a}-\sqrt{b}}$
 B. $\frac{a+2\sqrt{ab}-b}{a-b}$ D. $\frac{b+2\sqrt{ab}-a}{a-b}$
 C. $\frac{\sqrt{b}+\sqrt{a}}{2\sqrt{a}}$ E. $\frac{a+b}{a-b}$

5 If $3x^2 + ax - 5 \equiv (bx-1)(2-x) - 3$, then

- A. $a = -5, b = -3$
 B. $a = -5, b = 3$
 C. $a = -3, b = -5$
 D. $a = 5, b = -3$
 E. $a = 3, b = 5$

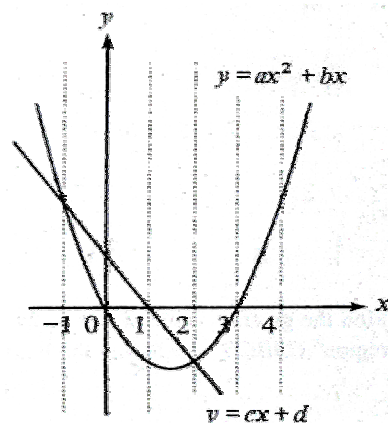
6 Find the greatest value of $3x+2y$ if (x, y) is a point lying in the region $OABCD$ (including the boundary).

- A. 3
 B. 4
 C. 2
 D. 9
 E. 8



7 The diagram shows the graphs of $y = ax^2 + bx$ and $y = cx + d$. The solutions of the equation $ax^2 + bx = cx + d$ are

- A. $-1, 1$
 B. $-1, 2$
 C. $0, 1$
 D. $0, 3$
 E. $1, 3$



8 If $\log(p+q) = \log p + \log q$, then

- A. $p = q = 1$
 B. $p = \frac{q}{q-1}$ D. $p = \frac{q+1}{q}$
 C. $p = \frac{q}{q+1}$ E. $p = \frac{q-1}{q}$

9 The expression $x^2 - 2x + k$ is divisible by $(x+1)$. Find the remainder when it is divided by $(x+5)$.

- A. 1
 B. 4 D. 16
 C. 12 E. 18

- 10 If 3, a , b , c , 23 are in A.S., then $a + b + c =$

A 13
B 26
C 33
D 39
E 65

- 11 Find the H.C.F. and L.C.M. of ab^2c and abc^3

H.C.F. L.C.M.

A a $a^2b^3c^4$
B aba ab^2c^3
C aba $a^2b^3c^4$
D ab^2c^3 aba
E $a^2b^3c^4$ aba

- 12 If α and β are the roots of the quadratic equation $x^2 - 3x - 1 = 0$, find the value of

$$\frac{1}{\alpha} + \frac{1}{\beta}$$

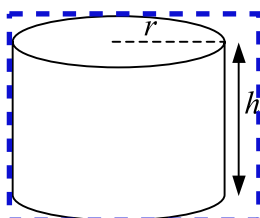
A -3
B -1
C $-\frac{1}{3}$
D $\frac{2}{3}$
E 3

- 13 If the simultaneous equations $\begin{cases} y = x^2 - k \\ y = x \end{cases}$ have only one solution, find k

A -1
B $-\frac{1}{4}$
C -4
D $\frac{1}{4}$
E 1

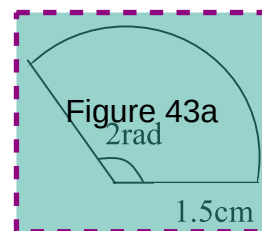
- 14 The price of a cylindrical cake of radius r and height h varies directly as the volume. If $r = 5$ cm and $h = 4$ cm, the price is \$30. Find the price when $r = 4$ cm and $h = 6$ cm.

A \$25
B \$28.80
C \$31.50
D \$36
E \$54



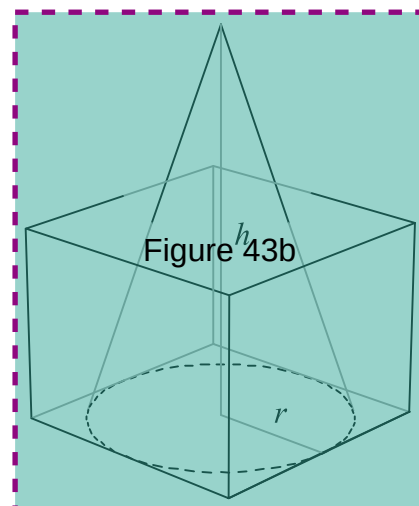
- 15 Find the perimeter of the sector in the figure.

A 2.25 cm
B 3 cm
C $\left(\frac{\pi}{60} + 3\right)$ cm
D 4.5 cm
E 6 cm



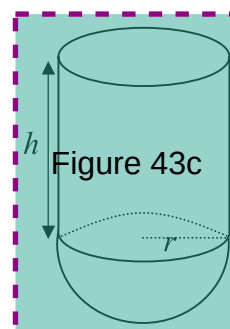
- 16 In the figure, the base of the conical vessel is inscribed in the bottom of the cubical box. If the box and the conical vessel have the same capacity, find $h : r$

A $24 : \pi$
B $3 : 1$
C $6 : \pi$
D $3 : \pi$
E $8 : 3\pi$



- 17 The figure shows a solid consisting of a cylinder of height h and a hemisphere of radius r . The area of the curved surface of the cylinder is twice that of the hemisphere. Find the ratio volume of cylinder : volume of hemisphere.

A 1 : 3
B 2 : 3
C 3 : 4
D 3 : 2



E. 3:1

- 18 A merchant marks his goods 25% above the cost. He allows 10% discount on the marked price for a cash sale. Find the percentage profit the merchant makes for a cash sale.

A. 12.5%
B. 15%
C. 22.5%
D. 35%
E. 37.5%

19 $\frac{\cos \theta}{1 - \sin^2 \theta} \times \frac{1 - \cos^2 \theta}{\sin \theta} =$

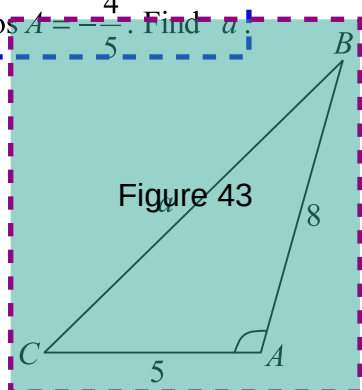
A. $\sin \theta$
B. $\cos \theta$
C. $\tan \theta$
D. $\frac{1}{\sin \theta}$
E. $\frac{1}{\cos \theta}$

20 $\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta =$

A. 0
B. 1
C. $(1 - \sin^2 \theta)^2$
D. $(1 - \cos^2 \theta)^2$
E. $(\cos^2 \theta - \sin^2 \theta)^2$

21 In the figure, $\cos A = -\frac{4}{5}$. Find a .

A. $\sqrt{153}$
B. $\sqrt{137}$
C. $\sqrt{89}$
D. $\sqrt{41}$
E. $\sqrt{25}$

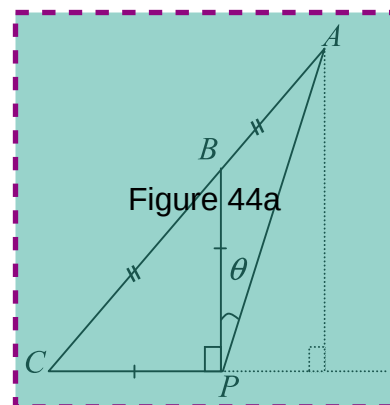


22 The largest value of $3 \sin^2 \theta + 2 \cos^2 \theta - 1$ is

A. 1
B. $\frac{3}{2}$
C. 3
D. 5

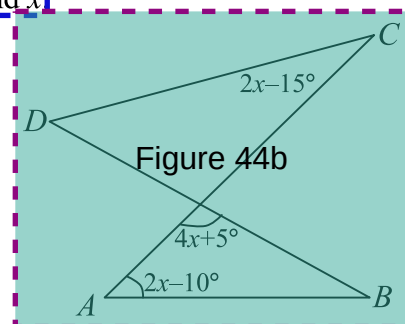
- 23 In the figure, $AB=BC$, $BP=CP$ and $BP \perp CP$. Find $\tan \theta$.

A. $\frac{1}{4}$
B. $\frac{1}{3}$
C. $\frac{1}{2}$
D. $\frac{1}{\sqrt{3}}$
E. $\frac{\sqrt{3}}{2}$



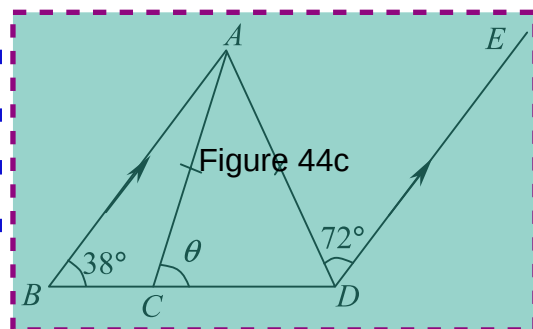
- 24 In the figure, points A, B, C and D are concyclic. Find x .

A. 20°
B. 22.5°
C. 25°
D. 27.5°
E. 30°



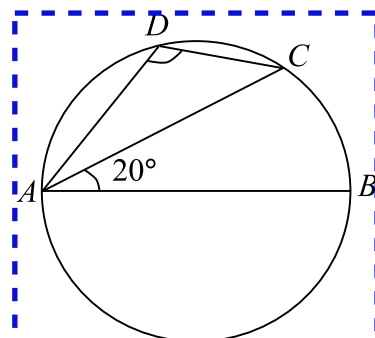
- 25 In the figure, $BA \parallel DE$ and $AC = AD$. Find θ .

A. 34°
B. 54°
C. 70°
D. 72°
E. 76°



- 26 In the figure, AB is a diameter. Find $\angle ADC$.

A. 100°
B. 110°
C. 120°
D. 135°
E. 140°



- 27 If the point $(1, 1)$, $(3, 2)$ and $(7, k)$ are on the same straight line, then $k =$

A. 3
B. 4
C. 6
D. 7
E. 10

- 28 $A(0, 0)$, $B(5, 0)$ and $C(2, 6)$ are the vertices of a triangle. $P(9, 5)$, $Q(6, 6)$ and $R(2, -9)$ are three points. Which of the following triangles has/have area(s) greater than the area of $\triangle ABC$?

I. $\triangle ABP$
II. $\triangle ABQ$
III. $\triangle ABR$

A. I only
B. II only
C. III only
D. I and II only
E. II and III only

- 29 A circle of radius 1 touches both the positive x -axis and the positive y -axis. Which of the following is/are true?

Label 45 I. Its center is in the first quadrant.

Label 11 II. Its center lies on the line $x - y = 0$.

Label 12 III. Its center lies on the line $x + y = 1$.

A. I only
B. II only
C. III only
D. I and II only
E. I and III only

- 30 What is the area of the circle $x^2 + y^2 - 10x + 6y - 2 = 0$?

A. 32π
B. 34π
C. 36π
D. 34π

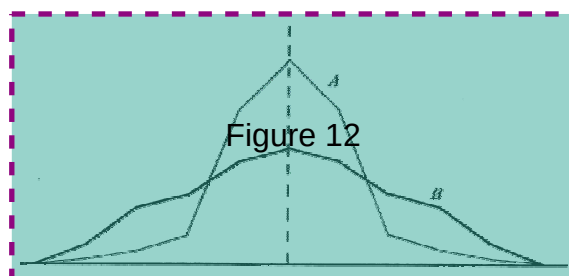
- 31 Two fair dice are thrown. What is the probability of getting a total of 5 or 10?

A. $\frac{1}{9}$
B. $\frac{5}{36}$
C. $\frac{1}{6}$
D. $\frac{7}{36}$
E. $\frac{2}{9}$

- 32 A group of n numbers has mean m . If the numbers 1, 2 and 6 are removed from the group, the mean of the remaining $n-3$ numbers remains unchanged. Find m .

A. 1
B. 2
C. 3
D. 6
E. $n-3$

33



The figure shows the frequency polygons of two symmetric distributions A and B with the same mean. Which of the following is/are true?

I. Interquartile range of $A <$ Interquartile range of B
II. Standard deviation of $A \geq$ Standard deviation of B
III. Mode of $A \geq$ Mode of B

A. I only
B. II only
C. III only
D. I and III only
E. II and III only

- 34 If $9^{x+2} = 36$, then $3^x =$

A. $\frac{2}{3}$

B. $\frac{4}{3}$

D. $\sqrt{6}$

C. 2

E. 9

35 If $a:b=2:3$ and $b:c=5:3$, then

$$\frac{a+b+c}{a-b+c} =$$

A. -2

B. $\frac{5}{2}$

D. $\frac{17}{2}$

C. 4

E. 3

36

x	Sign of $f(x)$
3.56	+
3.58	-
3.57	+
3.575	+

Figure 12b

From the table, a root of the equation $f(x)=0$ is

A. 3.57(correct to 3 sig.fig.)

B. 3.575(correct to 4 sig.fig.)

C. 3.5775(correct to 5 sig.fig.)

D. 3.5725(correct to 4 sig.fig.)

E. 3.58(correct to 3 sig.fig.)

37 Given that the positive numbers p, q, r, s are in G.S., which of the following must be true?

Label 18 kp, kq, kr, ks are in G.S., where k is a non-zero constant.

Label 11 a^p, a^q, a^r, a^s are in G.S., where a is a positive constant.

Label 12 $\log p, \log q, \log r, \log s$ are in A.S.

A. I only

B. II only

38 In the figure, the rectangle has perimeter 16cm and area 15cm^2 . Find the length of its diagonal

A.C.

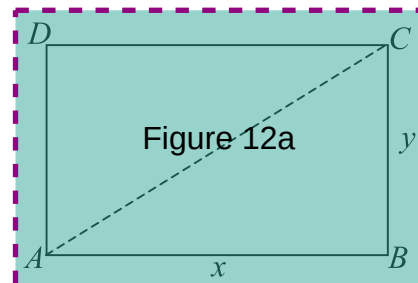
A. $\sqrt{32}\text{ cm}$

B. $\sqrt{34}\text{ cm}$

C. 7cm

D. $\sqrt{226}\text{ cm}$

E. $\sqrt{241}\text{ cm}$



39 In factorizing the expression $a^4 + a^2b^2 + b^4$, we find that

A. $(a^2 - b^2)$ is a factor.

B. $(a^2 + b^2)$ is a factor.

C. $(a^2 - ab - b^2)$ is a factor.

D. $(a^2 - ab + b^2)$ is a factor.

E. it cannot be factorized.

40 If the solution of the inequality $x^2 - ax + 6 \leq 0$ is $c \leq x \leq 3$, then

A. $a=5, c=2$

B. $a=-5, c=2$

C. $a=5, c=-2$

D. $a=1, c=-2$

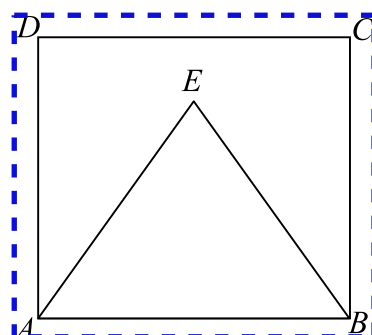
E. $a=-1, c=2$

41 In the figure, $ABCD$ is a square and ABE is an equilateral triangle. $\frac{\text{Area of } ABE}{\text{Area of } ABCD} =$

A. $\frac{1}{4}$

B. $\frac{1}{3}$

C. $\frac{\sqrt{3}}{8}$



D. $\frac{\sqrt{3}}{4}$

E. $\frac{\sqrt{3}}{2}$

- 42 In the figure, the radii of the sectors OPQ and ORS are 6cm and 3cm respectively.

$$\frac{\text{Area of shaded region}}{\text{Area of sector } OPQ} =$$

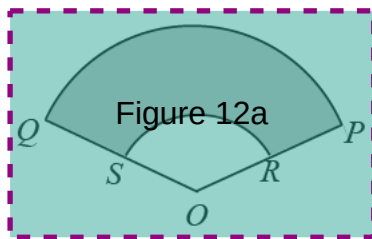
A. $\frac{4}{25}$

B. $\frac{2}{5}$

C. $\frac{9}{25}$

D. $\frac{16}{25}$

E. $\frac{21}{25}$



- 43 Which of the following gives the compound interest on \$10000 at 6% p.a. for one year, compounded monthly?

A. $\$10000 \times \frac{0.06}{12} \times 12$

B. $\$10000(1.06^{12} - 1)$

C. $\$10000 \left(1 + \frac{0.06}{12}\right)^{12}$

D. $\$10000 \left[\left(1 + \frac{0.06}{12}\right)^{12} - 1\right]$

E. $\$10000 \left[\left(1 + \frac{0.06}{12}\right)^{12} - 1\right]$

- 44 Originally $\frac{2}{3}$ of the students in a class failed in an examination. After taking a re-examination, 40% of the failed students passed. Find the total pass percentage of the class.

- 45 Solve $\tan^4 \theta + 2 \tan^2 \theta - 3 = 0$ for $0^\circ \leq \theta < 360^\circ$.

A. $45^\circ, 135^\circ$ only

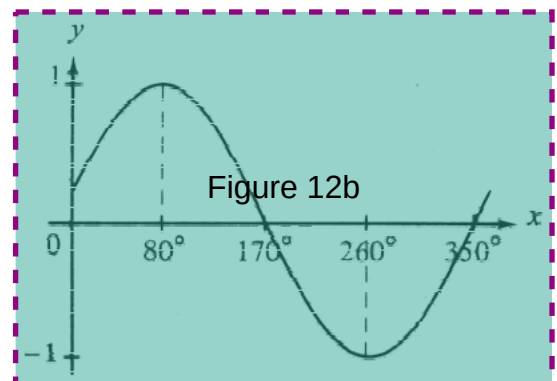
B. $45^\circ, 225^\circ$ only

C. $45^\circ, 60^\circ, 225^\circ, 240^\circ$

D. $45^\circ, 120^\circ, 225^\circ, 300^\circ$

E. $45^\circ, 135^\circ, 225^\circ, 315^\circ$

- 46 The figure shows the graph of the function



A. $y = \sin(350^\circ - x)$

B. $y = \sin(x + 10^\circ)$

D. $y = \sin(x - 10^\circ)$

C. $y = \cos(x + 10^\circ)$

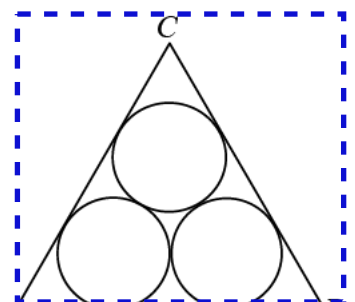
E. $y = \cos(x - 10^\circ)$

- 47 In the figure, ABC is an equilateral triangle and the radii of the three circles are each equal to 1. Find the perimeter of the triangle.

A. 12

B. $3(1 + \tan 30^\circ)$

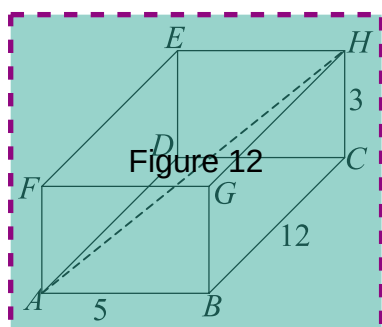
C. $6(1 + \tan 30^\circ)$



- D. $3\left(1 + \frac{1}{\tan 30^\circ}\right)$
- E. $6\left(1 + \frac{1}{\tan 30^\circ}\right)$

48 In the figure, $ABCDEFGH$ is a cuboid. The diagonal AH makes an angle θ with the base $ABCD$. Find $\tan \theta$.

- A. $\frac{3}{5}$
- B. $\frac{3}{12}$
- C. $\frac{3}{13}$
- D. $\frac{3}{\sqrt{178}}$
- E. $\frac{\sqrt{153}}{5}$



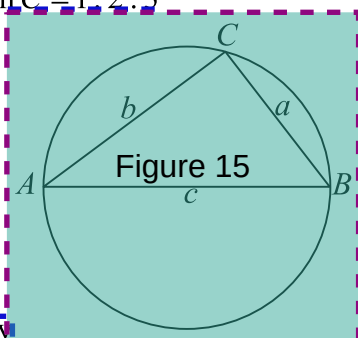
49 In the figure, if $\text{arc } BC : \text{arc } CA : \text{arc } AB = 1 : 2 : 3$, which of the following is/are true?

Label 13 $\angle A : \angle B : \angle C = 1 : 2 : 3$

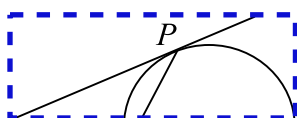
Label 14 $a : b : c = 1 : 2 : 3$

Label 15 $\sin A : \sin B : \sin C = 1 : 2 : 3$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I, II and III only



50 In the figure, TP and TQ are tangents to the circle at P and Q respectively. If M is a point on the minor arc PQ and $\angle PMQ = \theta$, then $\angle PTQ =$



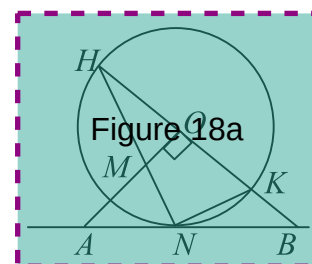
51 In the figure, O is the center of the circle. AB touches the circle at N . Which of the following is/are correct?

Label 16 M, N, K, O are concyclic.

Label 17 $\angle HNB \sim \angle NKB$

Label 18 $\angle OAN = \angle NOB$

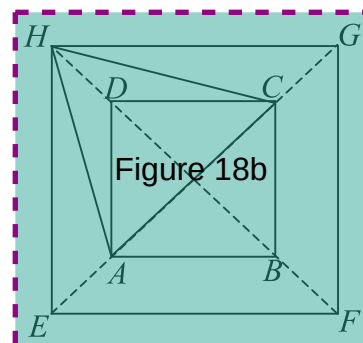
- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I, II and III



52 In the figure, $ABCD$ and $EFGH$ are two squares and ACH is an equilateral triangle.

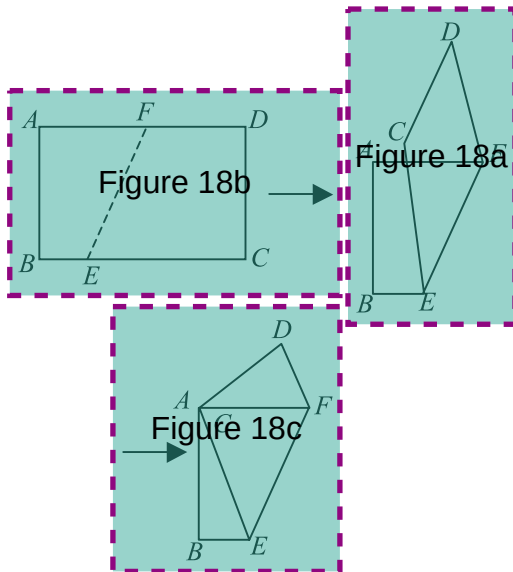
Find $AB : EE$.

- A. $1 : 2$
- B. $1 : 3$
- C. $1 : \sqrt{2}$
- D. $1 : \sqrt{3}$
- E. $\sqrt{2} : \sqrt{3}$



END OF PAPER

63



In the figure, a rectangular piece of paper $ABCD$ is folded along EF so that C and A coincide. If $AB = 12$ cm, $BC = 16$ cm, find BE .

A. 3.5 cm

B. 4.5 cm

C. 5 cm

D. 8 cm

E. 12.5 cm

64 In the figure, the three circles touch one another. XY is their common tangent. The two larger circles are equal. If the radius of the smaller circle is 4 cm, find the radii of the larger

Label 11
circles

A. 8 cm

B. 10 cm

C. 12 cm

